Pharmacotherapy of Bowel diseases

Third Stage Small Groups

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Pharmacotherapy of Inflammatory Bowel Disease (IBD)

IBD includes Crohn disease (CD) and ulcerative colitis (UC), which are characterized by:

Clinical manifestations

bloody diarrhea

mucus in stools

needing to pass stools urgently

cramping rectal pain, or tenesmus

malaise

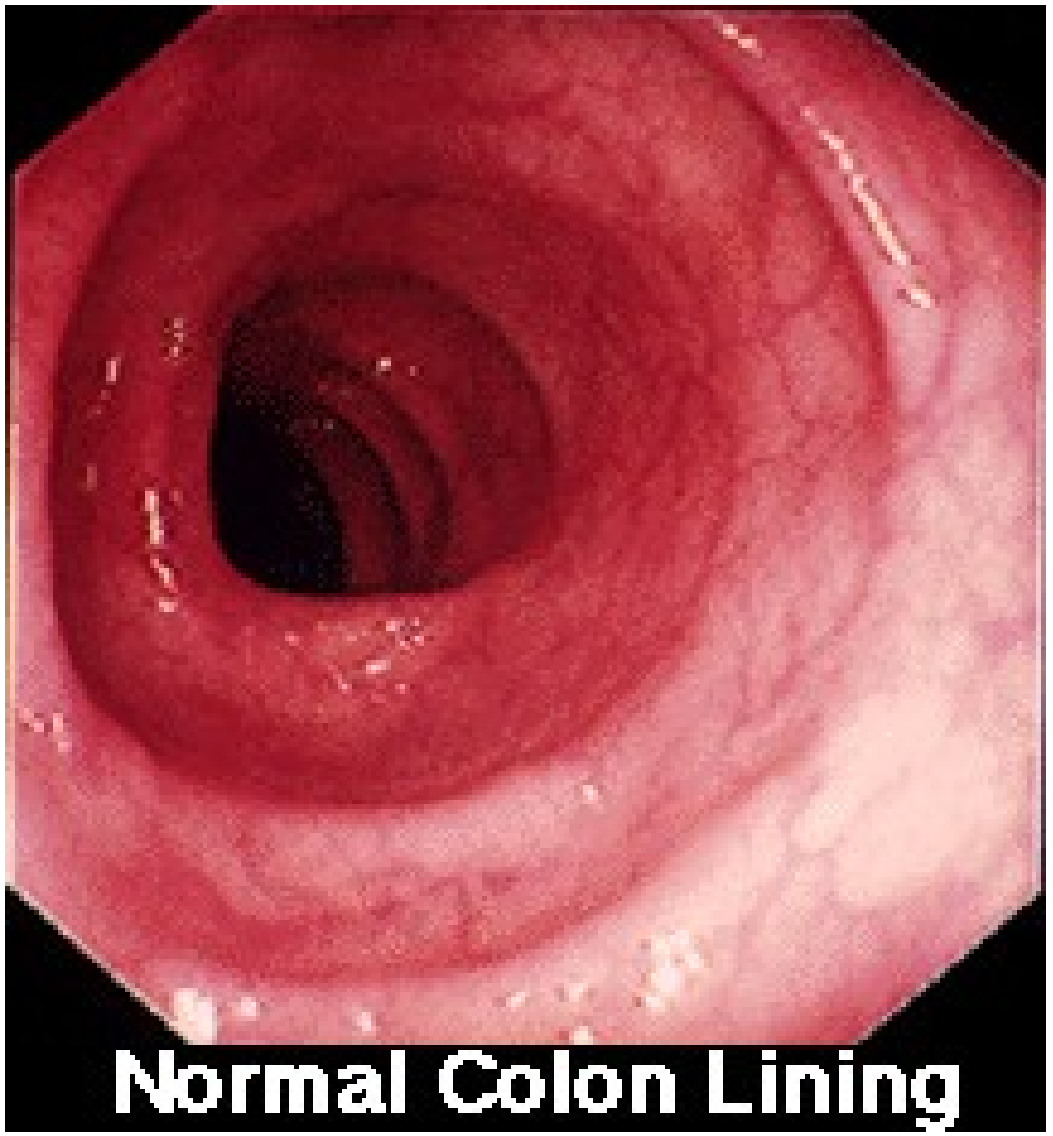
unexplained weight loss

fever

Clinical Features of UC and CD

Ulcerative Colitis	Crohn's Disease
Colon and rectum	Any part of the GI tract
Rectum universally involved	Rectum involved in 10% of cases
Mucosal and submucosal injury	Transmural injury that may lead to strictures or fistulae including perianal involvement
Continuous pattern of inflammation	Skip lesions
Acute onset	Insidious onset

~10% do not fit into either group and are deemed indeterminate colitis


Normal Colon Lining

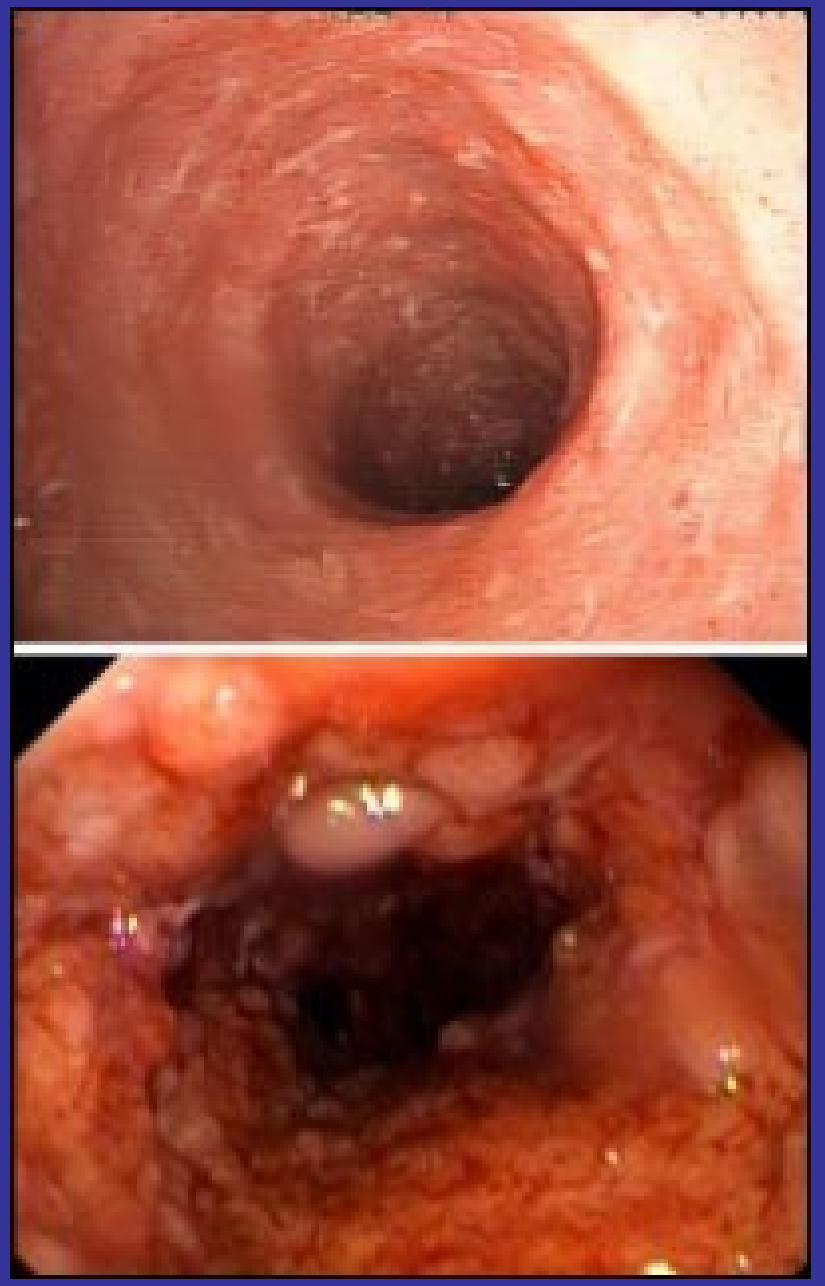

Ulcerative Colitis

Crohn’s disease

- patchy transmural inflammation

- fistulae, strictures

- any part of GI tract



COMMONLY USED DRUGS

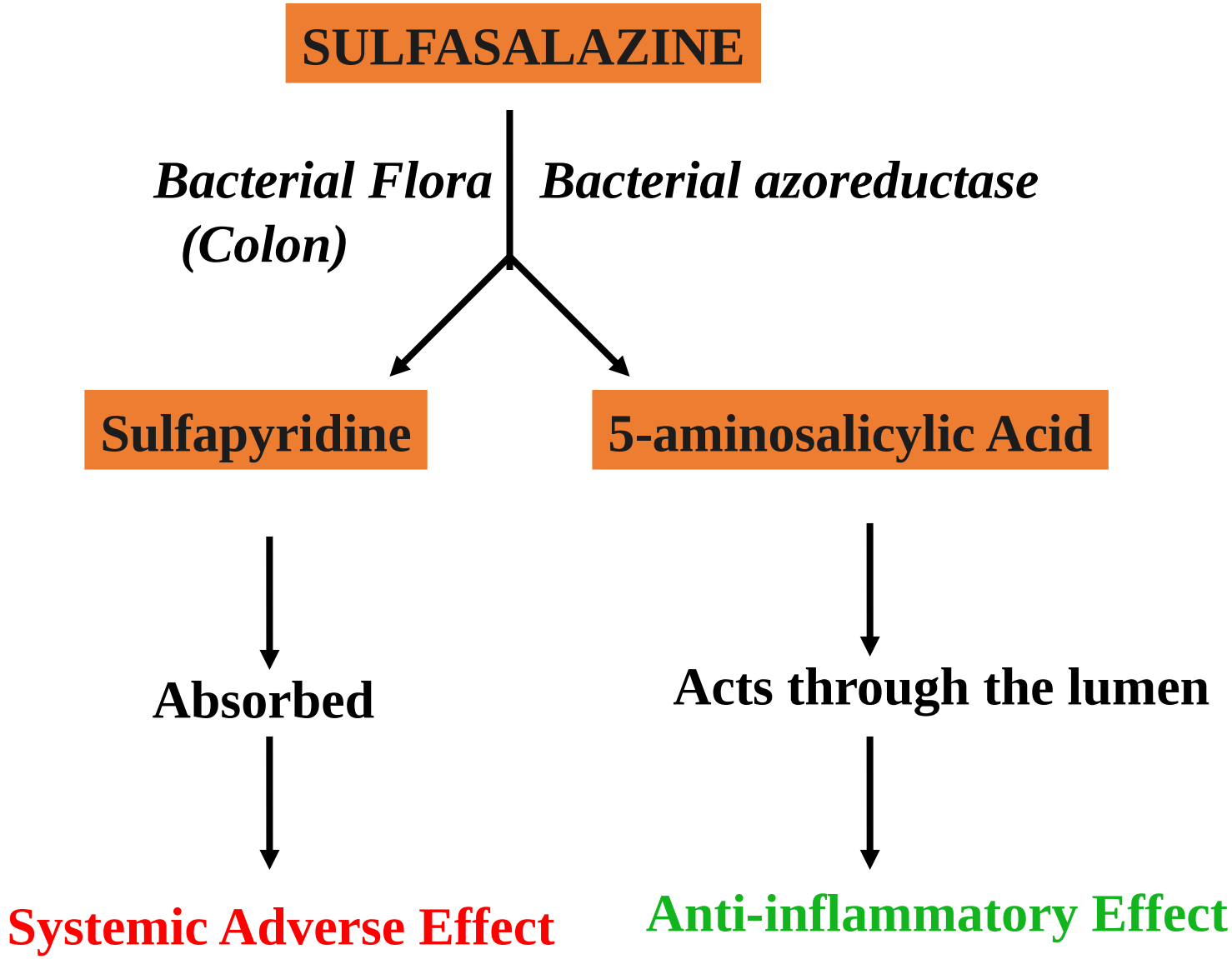
1. **Aminosalicylates:** Sulphasalazine, mesalamine, olsalazine, balsalazide.
2. **Glucocorticoids:** Prednisolone, methylprednisolone, hydrocortisone, budesonide.
3. **Immunomodulators**: Azathioprine, 6-mercaptopurine (6-MP), methotrexate, cyclosporine.
4. **. Biological response modifiers**: Infliximab, Adalimumab, Certolizumab, Vedolizumab and Ustekinumab
5. **JAK Inhibitors:** Tofacitinib (UC) and Upadacitinib for (UC and chrons).
6. **Antibiotics:** Metronidazole, ciprofloxacin, clarithromycin.
7. **Others:** Probiotics

Aminosalicylates

Sulphasalazine. It is a prodrug and is composed of **sulphapyridine and 5-aminosalicylic acid (5-ASA).** On oral administration, sulphasalazine reaches the colon, where it is broken down by colonic bacteria to 5-ASA and sulphapyridine. The released 5-ASA acts locally by inhibiting the production of inflammatory mediators. Sulphapyridine gets absorbed and causes side effects like nausea, vomiting, headache, bone arrow suppression and rarely oligospermia. Allergic side effects are skin rashes, fever, hepatitis, pancreatitis, pneumonitis, etc.

To avoid the side effects of sulphapyridine, several 5-ASA compounds have been developed which can be directly targeted to the colon.

It may reduce inflammation by scavenging free radicals, by inhibiting cyclooxygenase and lipooxygenase (inhibiting prostaglandin and leukotriene production), and/or by decreasing neutrophil chemotaxis and superoxide generation through decreased inflammatory cytokines (IL-1, TNF-α).


SULFASALAZINE
Bacterial Flora (Colon)
Bacterial azoreductase
Sulfapyridine
5-aminosalicylic Acid
Absorbed
Acts through the lumen
Systemic Adverse Effect
Anti-inflammatory Effect

Indications

- Inducing remission in mild UC/CD (higher doses)
- Maintaining remission in UC
- Less effective for maintenance in CD
- Reduce risk of colorectal cancer by 75% (long term Rx for extensive disease)

Mesalamine, Olsalazine and Balsalazide.

Mesalamine (mesalazine) is 5-ASA, well absorbed in the upper GIT; therefore, it has to be given as special formulations (delayed release capsules or pH-dependent tablets). It can be administered as suppository or enema.

Olsalazine is composed of two molecules of 5-ASA with an azo linkage, (a 5-ASA dimer linked by a bond that is hydrolysed by colonic bacteria) . It is poorly absorbed after oral administration. In the colon, it is cleaved into two molecules of 5-ASA by colonic bacteria.

Balsalazide, a prodrug from which 5 ASA is also released following hydrolysis of a diazo linkage and a metabolite in the colon.

Mesalamine, olsalazine and balsalazide have a lower incidence of side effects than sulphasalazine.

They may cause headache and skin rashes like **Steven's Johnson syndrome.**
Diarrhoea is common with olsalazine.
5-ASA agents are **mainly effective for mild to moderate ulcerative colitis**

Steven’s Johnson syndrome.



Glucocorticoids

Glucocorticoids are used for the **short-term treatment of moderate to severe IBD to induce remission** **not used for long term maintenance,** Oral and IV used for systemic diseases and rectal enema for distal colitis. Various glucocorticoids used in IBD are:

Prednisolone (oral),

Methylprednisolone (oral, parenteral),

Hydrocortisone (enema, suppository)

Budesonide (oral). High first pass metabolism with fewer systemic effects

Prolonged use of glucocorticoids can lead to hypothalamic–pituitary–adrenal axis suppression and other side effects like osteoporosis, peptic ulcer, infections and hyperglycaemia, hypertension with cushingoid features. MOA: Bind cytosolic glucocorticoid receptors leads to nuclear translocation and supessing transcription of pro-inflammatory genes (IL-2, TNF-α and INF-γ)

Immunosuppressants

Thiopurines: **Azathioprine, 6-mercaptopurine and Cyclosporine**

These drugs are used in **severe disease** or in **patients with steroid-dependent/steroid unresponsive IBD**.

MOA: converting to 6-thioguanine nucleotides inhibiting purine synthesis then blocking DNA/RNA synthesis in lymphocytes

PK notes: delayed onset 8-12 weeks

Metabolized by thiopurine methyltransferase (TPMT) inducing polymorphisms affect toxicity.

SE: myelosupression, hepatotoxicity, pancreatitis and increases the risk of lymphoma

Methotrexate: Given orally, IM and SC weekly to inhibits dihydrofolate reductase (low-dose effect leading to increase adenosine though decreasing inflammation. It is more effective in chrons rather than UC.

SE: bone marrow supression, hepatotoxicity and pulmonary fibrosis

Biological Response Modifiers

Anti- TNF agents : Infliximab, Adalimumab (TNF- inhibitors), Certolizumab and Golimumab.

Can be used in **severe cases of Crohn disease and refractory ulcerative colitis**. The main disadvantages of biologics are **their cost and increased susceptibility to infections.**

MOA: bind TNF-α leading to prevent interaction with TNFR then inhibiting pro-inflammatory signaling.

SE: infusion reaction, infections (especialy TB, Hepatitis-B reactivatiobn demyelination and Lymphoma

Anti Integrin: Vedolizumab (α4β7 integrin antibody)

MOA: blocks lymphocyte trafficking

SE: Safer systemic profile, rare PML risk (much lower than natalizumab)

Anti IL-12/23: Ustekinumab used for moderate to sever chrons and UC

MOA: binds p40 subunit of IL-12 and IL-23 inhibiting Th1/Th17 pathways

SE: Infections and rarely malignancy.

JAK Inhibitors (Small molecules) given orally

Examples : **Tofacitinib (UC) and Upadacitinib (UC and chrons disease)**

MOA: inhibits Janus Kinases (JK1/3 mainly then blocks STAT signaling and inhibiting cytokine driven inflammation

SE: Infections (herpes zoster, hyperlipidemia, cytopenias and rarely thrombaz

sphingosine-1-phosphate (S1P) receptors.

They work primarily by **reducing the movement of lymphocytes (immune cells) from lymph nodes into the bloodstream**, thereby decreasing immune-system activity and inflammation. **Examples include:**

- **Ozanimod**
- **Fingolimod**
- **Siponimod**
- **Ponesimod**
- **Etrasimod**

They are used to treat certain **immune-mediated and inflammatory diseases**, including **multiple sclerosis (MS)** and, depending on the specific drug, **ulcerative colitis**.

Ozanimod (Zeposia)

Ozanimod is an oral **sphingosine-1-phosphate (S1P) receptor modulator**. It selectively targets **$S1P_1$ and $S1P_5$ receptors**.

Main mechanism: Ozanimod causes certain lymphocytes to remain within lymph nodes, reducing their circulation to the central nervous system and other tissues. This decreases inflammation and abnormal immune activity.

Main approved uses:

- **Relapsing forms of multiple sclerosis (MS)**
- **Moderate-to-severe ulcerative colitis** in adults

Drug class: S1P receptor modulator

Route and Dosage: Oral 0.23 mg once daily increase to 0.92mg after one week

Important considerations include **bradycardia/heart-rate effects, liver function, blood pressure, infections, macular edema, and drug interactions**. It also requires specific assessment before initiation in some patients.

Probiotics

Are live microorganisms (mostly bacteria and some yeasts) that, when administered in adequate amounts, confer a health benefit on the host, particularly by improving or restoring gut flora.

Lactobacillus species , Streptococcus thermophilus , Bifidobacterium species, Saccharomyces boulardii (yeast used for diarrhea)

Probiotic Formulations: are available as:

Capsules/tablets
Sachets
Yogurt and fermented milk
Fortified foods (e.g., probiotic drinks)
Synbiotics: Probiotics + Prebiotics (e.g., inulin)

Examples of Probiotic Products (India & globally) :**Sporlac, Darolac, Yakult (drink), Florastor** (S. boulardii)

Therapeutic uses in GIT

- Antibiotic-associated diarrhea (AAD)
- **Infectious diarrhea** (viral or bacterial
- Traveler’s diarrhea
- Irritable Bowel Syndrome (IBS)
- IBD
- Lactose intolerance
- Constipation
- Infant colic
- H. Pylori infection

- Restore gut flora, reduce risk of *Clostridium difficile* infection
- Shorten duration and severity
- Prevention and treatment
- Reduce bloating, flatulence, and improve stool regularity
- Adjunctive therapy in *ulcerative colitis* and *pouchitis*
- Aid in lactose digestion via bacterial lactase
- Some strains improve stool frequency and consistency
- Certain strains like *L. reuteri* may help
- As adjuvants to triple therapy (may reduce side effects and improve eradication rates)

Considerations and Safety

Generally safe, especially in healthy individuals

Use with caution in:

- Immunocompromised patients
- Critically ill or ICU patients
- Patients with central venous catheters (risk of sepsis)

During a Crohn’s flare-up, it’s important to avoid foods that can irritate the gut. A diet called the Crohn’s disease exclusion diet (CDED) can help. The diet may lower your symptoms and support healing.

Cut out foods

- Wheat and other gluten-containing grains
- Dairy products
- Animal fats
- Additives and ultra-processed foods
- Red meat

Encourage foods

- Whole foods
- Fruits
- Vegetables
- Rice
- Potatoes
- Lean meats and eggs

Pharmacological therapies for irritable bowel syndrome

Irritable bowel syndrome (IBS)

Is a prevalent functional gastrointestinal disorder, which represents a major cost to healthcare services.

Current pharmacological treatment includes fibre supplements, antispasmodics, laxatives, loperamid and antidepressants.

The novel pharmacological treatments already or recently approved for patients with IBS-C (lubiprostone, linaclotide) and IBS-D (alosetron, ramosetron, rifaximin, eluxadoline).

Other idrugs, with potential application in IBS (prucalopride, elobixibat, mesalazine, ondansetron and colesevelam)

Eluxadoline is a mixed μ and κ opioid-receptor agonist and δ-receptor antagonist that has recently been licensed for treatment of IBS with diarrhea. The drug acts on opioids receptors in enteric neurons that regulate motility and visceral sensation in the GI tract, resulting in slowing of intestinal transit and improved stool consistency.

Eluxadoline has low oral bioavailability and is considered to have limited potential for adverse effects on opioid receptors in the central nervous system.

Linaclotide has recently received regulatory approval for symptomatic treatment of moderate to severe IBS with constipation in adults. It is a synthetic peptide that is structurally related to endogenous guanylin peptides. Linaclotide is an agonist at the guanylate cyclase-C receptor on the luminal surface of intestinal epithelium, and increases the concentration of cyclic guanosine monophosphate in the intestinal cells. This results in greater secretion of chloride and bicarbonate ions and intestinal fluid, as well as more rapid intestinal transit. Clinical trials have demonstrated improvements in bowel movements and reduction in abdominal discomfort, although diarrhoea is a recognised adverse effect

For Irritable Bowel Syndrome with Constipation (IBS-C)

Linaclotide (Linzess): This prescription medication increases fluid secretion in the small intestine to promote more frequent and easier bowel movements. It can also help reduce abdominal pain and bloating.

Lubiprostone (Amitiza): A prescription drug that increases fluid secretion in the intestines. It is approved for women with IBS-C, as trials initially enrolled a limited number of men, though it has been effective in both.

Plecanatide (Trulance): This drug works similarly to linaclotide by increasing fluid in the intestines to soften stools and speed transit time. It also helps relieve abdominal pain.

Tenapanor (Ibsrela): This drug increases water secretion into the intestines by blocking sodium absorption, which helps soften stool and accelerates intestinal transit.

Tegaserod (Zelnorm): This medication was reintroduced to the market under specific restrictions due to previous cardiovascular safety concerns. It is approved for women under 65 with IBS-C who have not responded to other treatments and are not at risk for cardiovascular disease.

For Irritable Bowel Syndrome with Diarrhea (IBS-D)

Rifaximin (Xifaxan): An antibiotic that can decrease bacterial overgrowth in the gut, which some believe is a cause of IBS-D. A two-week course can provide lasting symptom relief for some patients.

Eluxadoline (Viberzi): A controlled substance that works on opioid receptors in the digestive system to reduce bowel contractions and fluid secretion, relieving diarrhea and abdominal pain.

Alosetron (Lotronex): This drug slows the movement of waste through the colon. It is reserved for women with severe IBS-D who have not responded to other treatments due to rare but serious side effects.

Loperamide (Imodium): This OTC antidiarrheal can help control diarrhea and slow intestinal contractions. It is not proven to relieve abdominal pain and is typically for short-term use.

Bile acid binders: Drugs such as cholestyramine can help if excessive bile acids in the colon are causing diarrhea.

For pain and bloating in IBS

Antispasmodics: These drugs relax the smooth muscles of the intestines to relieve painful spasms and cramping. Options include dicyclomine (Bentyl) and hyoscyamine (Levsin).

Peppermint oil: Enteric-coated capsules of peppermint oil can act as an antispasmodic by relaxing intestinal muscles. It can be particularly effective for pain and bloating.

Neuromodulators (Antidepressants): Tricyclic antidepressants (TCAs): Drugs like amitriptyline (Elavil) are often used in low doses to relieve pain and normalize gut motility by affecting nerve signals between the gut and brain. They are often used for diarrhea-predominant IBS, as they can cause constipation.

Selective serotonin reuptake inhibitors (SSRIs): These antidepressants, such as fluoxetine (Prozac), can be used to treat IBS, especially when anxiety or depression are also present. Their effect on motility may benefit those with IBS-C.

Other pain medications: Some studies have shown that nerve pain medications like **pregabalin** (Lyrica) or **gabapentin** (Neurontin) may help relieve severe abdominal pain and bloating in people with IBS.

Drugs for Acid-Peptic Disorders

These help in reducing or neutralizing stomach acid.

a. Antacids

- Aluminum hydroxide
- Magnesium hydroxide
- Calcium carbonate
- Sodium bicarbonate

b. H2 Receptor Blockers

- Ranitidine *(less used due to safety concerns)*
- Famotidine
- Nizatidine

c. Proton Pump Inhibitors (PPIs)

- Omeprazole
- Pantoprazole
- Esomeprazole
- Lansoprazole

d. Mucosal Protective Agents

- Sucralfate
- Bismuth subsalicylate
- Misoprostol *(also a prostaglandin analog)*